

Cybersecurity Home Lab Setup & Documentation

A hands-on cybersecurity lab environment for practising real-world attack & defence techniques

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Tools: Nmap | Wireshark | Burp Suite | Kali Linux | Windows Server

Year: 2024 | Type: Personal Security Research Project

1. Project Overview

This project documents the design, setup, and usage of a personal cybersecurity home lab built to simulate real-world enterprise network environments. The goal was to gain practical, hands-on experience in network security, penetration testing, and vulnerability assessment — skills that go beyond theoretical knowledge.

Objective	Description
Network Simulation	Simulate an enterprise-like network with multiple machines
Penetration Testing	Practice ethical hacking tools in a safe, isolated environment
Traffic Analysis	Capture and analyse real network packets using Wireshark
Vulnerability Assessment	Identify open ports, services, and misconfigurations using Nmap
Reporting	Document findings with risk levels and remediation steps

2. Lab Environment Setup

2.1 Hardware & Virtualization

The home lab was built using virtualization software to run multiple operating systems simultaneously on a single physical machine. This approach is cost-effective and allows for quick snapshots and rollbacks.

Component	Details
Host Machine	Personal Laptop / Desktop
Virtualization	VirtualBox / VMware Workstation
Attacker Machine	Kali Linux (primary attack platform)
Target Machine 1	Windows Server (Active Directory environment)
Target Machine 2	Windows 10 (end-user simulation)
Network Mode	Host-Only / NAT Network (isolated from internet)

2.2 Network Architecture

The lab network was configured as an isolated environment to ensure no unintended traffic reaches the internet:

- Kali Linux assigned IP: 192.168.56.10 (Attacker)

- Windows Server assigned IP: 192.168.56.20 (Target/Domain Controller)
 - Windows 10 Client assigned IP: 192.168.56.30 (Target/End User)
 - All machines on Host-Only network — completely isolated from external internet
- *All testing was performed in this isolated lab environment. No real systems were targeted.*

3. Network Scanning with Nmap

3.1 What is Nmap?

Nmap (Network Mapper) is an open-source tool used for network discovery and security auditing. It helps identify live hosts, open ports, running services, and potential vulnerabilities on a network.

3.2 Scans Performed

Basic Host Discovery

```
$ nmap -sn 192.168.56.0/24
```

Purpose: Identify all live hosts on the lab network.

Result: Discovered 3 active hosts — .10 (Kali), .20 (Windows Server), .30 (Win10)

Full Port Scan

```
$ nmap -sV -sC -p- 192.168.56.20
```

Purpose: Detect all open ports and running services on Windows Server.

Port	State	Service	Version	Risk
22	Open	SSH	OpenSSH 7.9	Low
80	Open	HTTP	Apache 2.4	Medium
135	Open	MSRPC	Microsoft RPC	Medium
445	Open	SMB	Windows SMB	High
3389	Open	RDP	Microsoft RDP	High
389	Open	LDAP	Active Directory	Medium

■ *Ports 445 (SMB) and 3389 (RDP) flagged as HIGH risk — commonly exploited in real attacks.*

4. Network Traffic Analysis with Wireshark

4.1 What is Wireshark?

Wireshark is a network protocol analyser that captures and inspects packets travelling across a network in real time. It is widely used by security professionals to detect anomalies, analyse attacks, and troubleshoot network issues.

4.2 Analysis Performed

- Captured live traffic between Kali Linux and Windows Server on the lab network
- Applied filters to isolate specific protocols: HTTP, DNS, SMB, ARP, ICMP
- Detected cleartext credentials being passed over HTTP (demonstrating importance of HTTPS)

- Observed ARP requests to map IP-to-MAC address relationships on the network
- Analysed DNS queries to understand how domain name resolution works in the lab

4.3 Key Wireshark Filters Used

Filter	Purpose
http	Show all HTTP traffic
dns	Show all DNS queries and responses
smb smb2	Show SMB file sharing traffic
tcp.port == 3389	Isolate RDP traffic
arp	View ARP requests/replies
ip.addr == 192.168.56.20	Filter traffic to/from Windows Server

5. Vulnerability Assessment Findings

Based on the Nmap scans and Wireshark traffic analysis, the following vulnerabilities and misconfigurations were identified in the lab environment:

#	Finding	Severity	Recommendation
1	RDP (3389) Exposed	HIGH	Restrict RDP access to specific IPs only
2	SMB (445) Open	HIGH	Disable SMBv1, use SMBv3 with encryption
3	HTTP Cleartext Login	HIGH	Enforce HTTPS with SSL/TLS certificates
4	SSH Default Config	MEDIUM	Disable root login, use key-based auth
5	RPC Port Exposed	MEDIUM	Restrict MSRPC with firewall rules
6	LDAP Unencrypted	MEDIUM	Enable LDAPS (LDAP over SSL) on port 636

6. Skills Gained

Technical Skills

- Network scanning and enumeration using Nmap
- Packet capture and protocol analysis using Wireshark
- Understanding of TCP/IP, DNS, SMB, RDP, LDAP protocols
- Identifying open ports and assessing their risk levels
- Documenting vulnerabilities and writing remediation recommendations
- Setting up and managing VMs for isolated lab environments

Soft Skills

- Structured thinking and systematic approach to security testing
- Technical documentation and professional report writing
- Self-directed learning and problem-solving

7. Disclaimer

All activities documented in this report were performed exclusively within a privately owned, isolated virtual lab environment. No external systems, networks, or third-party infrastructure were targeted or accessed at any point. This project was created purely for educational purposes and personal skill development in the field of cybersecurity. The author does not condone or support any form of unauthorized access to computer systems or networks.
